

AKSHA, Russian Federation

WMO: 309570

Lat: 50.267N

Lon: 113.267E

Elev: 732

StdP: 92.83

Time Zone: 9.00 (E09)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-35.6	-33.2	-38.0	0.1	-35.0	-35.6	0.1	-32.6	9.3	-13.9	8.0	-14.6	1.3	190	0.655

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.9	31.0	17.3	28.8	16.8	26.7	16.2	19.6	26.3	18.7	25.0	17.8	23.8	3.6	210

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	13.9	21.6	16.7	13.0	20.7	15.7	12.2	19.9	59.6	26.6	56.4	24.9	53.4	23.7	24.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.5	8.2	7.2	-39.9	34.8	4.2	2.9	-42.9	36.8	-45.4	38.5	-47.7	40.1	-50.8	42.2		
(5)			-39.2	21.7	3.5	1.2	-41.7	22.6	-43.7	23.2	-45.7	23.9	-48.2	24.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	-0.1	-21.7	-17.0	-7.5	2.2	9.5	16.6	19.0	16.8	9.7	0.4	-11.0	-19.6
	DBStd	15.02	5.22	5.94	6.34	4.56	4.03	3.63	2.75	3.31	3.99	4.80	6.25	5.76
	HDD10.0	4475	982	756	542	236	60	2	0	1	53	297	629	917
	HDD18.3	6823	1240	989	801	483	276	74	22	69	259	555	879	1176
	CDD10.0	781	0	0	0	3	43	198	281	211	44	1	0	0
	CDD18.3	87	0	0	0	0	1	21	44	21	1	0	0	0
	CDH23.3	1444	0	0	0	4	72	453	585	301	29	0	0	0
	CDH26.7	472	0	0	0	1	18	148	207	93	4	0	0	0
Wind	WSAvg	2.6	2.3	2.4	2.8	3.4	3.6	2.7	2.3	2.2	2.5	2.6	2.4	2.2
Precipitation	PrecAvg	386	2	2	4	13	32	52	113	99	44	9	4	4
	PrecMax	632	7	11	16	40	87	119	264	298	138	32	18	14
	PrecMin	235	0	0	0	0	3	3	30	24	2	0	0	0
	PrecStd	106	2	3	4	11	23	28	47	55	30	8	5	3
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-4.6	2.9	13.0	22.1	29.5	33.4	35.2	32.2	26.9	19.0	7.2	-2.7
		MCWB	-7.0	-1.9	4.6	9.6	14.0	17.5	18.9	18.3	14.8	8.8	1.7	-5.2
	2%	DB	-8.0	-0.9	9.1	17.9	25.5	30.4	31.4	29.2	23.3	14.7	3.5	-5.8
		MCWB	-9.7	-4.3	2.3	7.6	12.4	16.5	18.1	17.6	13.1	6.6	-0.5	-7.8
	5%	DB	-10.7	-3.8	6.0	14.8	22.5	28.1	28.7	26.6	20.6	11.3	0.9	-8.4
		MCWB	-12.1	-6.5	0.4	5.9	10.9	15.9	17.7	17.1	11.8	4.8	-2.2	-9.8
	10%	DB	-12.9	-6.5	3.0	11.9	19.7	25.9	26.3	24.2	18.0	8.7	-1.4	-10.7
		MCWB	-14.0	-8.5	-1.5	4.2	9.4	15.1	17.3	16.4	10.7	3.2	-3.9	-11.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-6.9	-1.6	5.2	10.2	14.7	19.7	21.1	20.6	16.2	9.1	1.7	-5.0
		MCDB	-4.6	2.7	13.1	21.0	27.0	27.5	27.9	28.3	23.9	18.6	6.7	-2.9
	2%	WB	-9.7	-4.3	2.5	7.9	13.0	18.2	20.0	19.1	14.4	6.8	-0.4	-7.6
		MCDB	-8.0	-1.1	8.7	17.4	24.3	25.8	27.0	25.1	20.2	13.7	3.2	-5.8
	5%	WB	-12.0	-6.4	0.5	6.1	11.5	17.1	19.2	18.1	13.0	5.1	-2.2	-9.8
		MCDB	-10.8	-3.8	5.8	14.3	20.9	24.5	25.6	24.1	18.4	10.6	0.8	-8.4
	10%	WB	-14.0	-8.5	-1.3	4.5	10.1	16.2	18.3	17.2	11.6	3.5	-3.9	-11.8
		MCDB	-13.0	-6.5	2.9	11.4	18.7	23.6	24.1	22.7	17.0	8.2	-1.4	-10.7
Mean Daily Temperature Range	5% DB	MDBR	13.0	16.2	15.7	14.3	15.5	14.9	12.9	13.0	13.9	12.9	12.1	11.7
		MCDBR	14.2	18.3	18.5	20.3	22.3	20.3	18.3	17.8	19.0	18.1	13.5	12.5
		MCWBR	12.6	15.1	12.2	11.6	10.9	8.3	7.1	7.6	9.4	10.6	9.9	10.8
	5% WB	MCDBR	14.0	18.1	17.9	19.1	19.7	15.8	13.7	14.1	15.8	17.0	13.3	12.5
		MCWBR	12.5	15.1	11.9	11.2	9.9	7.2	6.3	6.8	8.0	10.2	9.8	10.8
Clear-Sky Solar Irradiance	taub	0.249	0.274	0.313	0.380	0.457	0.447	0.419	0.381	0.334	0.299	0.270	0.254	
	taud	2.297	2.271	2.263	2.171	1.978	2.130	2.273	2.362	2.435	2.451	2.367	2.256	
	Ebn at Noon	804	869	890	859	801	815	834	852	861	834	770	729	
	Edh at Noon	66	89	108	134	171	148	126	110	92	74	62	59	
All-Sky Solar Radiation	RadAvg	1.63	2.77	4.33	5.39	5.96	6.12	5.63	5.01	4.04	2.70	1.65	1.25	
	RadStd	0.05	0.09	0.14	0.39	0.34	0.41	0.43	0.38	0.29	0.14	0.09	0.08	

Historical Trends

		Heating		Cooling			Degree-Days				
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (8 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page